

CHITHMI PEIRIS

BSc (Hons) Plant Biotechnology - First Class

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PROFILE

First Class Honours graduate in Plant Biotechnology from Sri Lanka, currently pursuing an MPhil in Microbiology with research focused on microbial secondary metabolites, natural pigments, and marine biotechnology. Experienced in microbial isolation, fermentation optimization, molecular characterization, chromatographic analysis, and spectroscopy. Passionate about developing sustainable microbial products for industrial and biomedical applications through interdisciplinary research.

EDUCATION

Master of Philosophy (MPhil) in Microbiology (Research)

2024 - Present (Expected completion: December 2026)

University of Sri Jayewardenepura, Sri Lanka.

BSc Honors Degree in Plant Biotechnology

2020 - 2024

University of Colombo, Sri Lanka.

Overall GPA: 3.73 / 4.00, Major GPA: 4.00 / 4.00 (First Class Division)

Key areas: Microbiology, Molecular Biology, Plant Biochemistry, Biostatistics, Bioinformatics, Phylogenetic Analysis, Plant Pathology, Bioprospecting, Post-Harvest Technology, Agrobiotechnology, and Horticulture.

G.C.E. Advanced Level Examination

2018

Taxila Central College, Horana, Sri Lanka.

Biological Science stream: Chemistry A, Biology A, Physics C

G.C.E. Ordinary Level Examination

2015

Panadura Balika Vidyalaya, Panadura, Sri Lanka.

Nine High Distinctions (9As).

RESEARCH EXPERIENCE

1. Undergraduate Research Project

2023 - 2024

Department of Plant Sciences, University of Colombo.

Project Title - "Extraction and Characterization of dyes of fungal origin"

Research supervisors:

Dr. Jinendra Dissanayake (Senior Lecturer, Department of Plant Sciences, University of Colombo)

Dr. Sameera Ariyawansa (Senior Lecturer, Department of Plant Sciences, University of Colombo)

Prof. M.N. Kaumal (Professor, Department of Chemistry, University of Colombo)

- Screened previously preserved fungal isolates from lichen (*Rocella montagne*) samples collected at the Kalpitiya Lagoon, Sri Lanka to identify pigment-producing species (*Talaromyces verruculosus*).
- Optimized submerged Liquid state fermentation conditions.
- Extracted pigments using ethyl acetate, and purified by Thin Layer Chromatography (TLC) and Characterized compounds using UV-Vis, FTIR, and NMR.
- Identified pigment as austrocortirubin (anthraquinone polyketide), contributing to sustainable alternatives for synthetic dyes.

2. Independent Research-Based Undergraduate Assignment

2023 - 2024

Department of Plant Sciences, University of Colombo.

Project Title – “Trends in production and consumption of herbal apparel in Sri Lanka.”

Research supervisors:

Dr. Jinendra Dissanayake

3. Undergraduate Group Research Project

May 2022 - November 2022

Department of Plant Sciences, University of Colombo.

Project Title – “Evaluation of protocols for the extraction of high-quality genomic DNA from black pepper leaves (*Piper nigrum* L.)”

Research supervisors:

Prof. H.D. Dharshani Bandupriya (Professor, Department of Plant Sciences, University of Colombo)

Dr. Anushka Wikramasuriya (Senior Lecturer, Department of Plant Sciences, University of Colombo)

PROFESSIONAL EXPERIENCES

1. Graduate Research Assistant

January 2025 - Present

Department of Botany, University of Sri Jayewardenepura, Sri Lanka.

Ongoing Research Project Title: “Investigation of marine pigmented bacteria associated with the phycoplankton of brown seaweeds for UV blocking and antioxidant properties”.

Research supervisor – Dr. Isuru U. Kariyawasam (Senior Lecturer, Department of Botany, University of Sri Jayewardenepura)

- Isolation, Screening, and cultivation of marine pigmented bacterial strains.
- Optimization of fermentation parameters for bacterial pigment production.
- Pigment extraction, purification, and analysing using chromatographic techniques.
- Performed antioxidant assays (DPPH, ABTS, and FRAP) to determine the bio-active potential.
- Molecular characterization of bacterial isolates through DNA extraction and 16S rRNA PCR amplification.

2. Graduate Teaching Assistant

May 2024 - January 2025

Department of Plant Sciences, University of Colombo, Sri Lanka.

Conducted and assisted practical sessions for undergraduate and postgraduate courses.

Assisted in examination supervision and invigilation.

Courses taught:

- Genetics and Cell Biology (Level I)

- Plant Pathology (Level III)
- Applied Microbiology (Level III)
- Experimental Design and Data Analysis (Level III)
- Horticulture (Level III)
- Environment & Biodiversity Related Legislation in Sri Lanka (Level III)
- Food Microbiology (MSc in Agricultural Microbiology)

PUBLICATIONS

Book Chapters

1. Peiris, K.C.T., & Kariyawasam, I.U. (2025). From ocean to industry: Novel bio-innovations with marine bacterial pigments. In *Biological Innovations for Environmental Sustainability* (pp. 309–324). Colombo: Institute of Biology, Sri Lanka.
2. Peiris, K.C.T., Dissanayake, I.A.J.K., & Ariyawansa, K.G.S.U. (2024). Dyes of fungal origin: Reshaping the dye industry with sustainable alternatives to synthetic dyes. In *Achieving a Circular and Sustainable Bioeconomy* (pp. 309–324). Colombo: Institute of Biology, Sri Lanka.

Peer-Reviewed / Preprint Articles

3. Peiris, K.C.T., Dissanayake, I.A.J.K., & Ariyawansa, K.G.S.U. (2024). Extraction and characterization of dyes of fungal origin. *SSRN Electronic Journal* (Preprint). Elsevier BV. <https://doi.org/10.2139/ssrn.> Explores sustainable fungal-derived alternatives to synthetic dyes, providing novel insights for eco-friendly industrial dye production.

Conference Papers

4. Peiris, K.C.T., Kariyawasam, I.U., Sanjeewa, K.K.A., & Ambagaspitiya, S. (2026). Antioxidant and UV-protective potential of pigmented *Brevibacterium* sp. isolated from *Padina* sp., collected in Hikkaduwa, Sri Lanka. Oral presentation at the *International Conference on Marine Science and Sustainability (ICMSS) 2026*, Ministry of Environment, Colombo, Sri Lanka.
5. Hemachandra, P.C.L., Peiris, K.C.T., Kariyawasam, I.U., & Pathirana, E. (2026). Characterization of Biosurfactant-Producing Bacteria Associated with Brown Macroalgae from the Southern Coast of Sri Lanka. Oral presentation at the *International Conference on Marine Science and Sustainability (ICMSS) 2026*, Ministry of Environment, Colombo, Sri Lanka.
6. Peiris, K.C.T., Kariyawasam, I.U., Sanjeewa, K.K.A., & Ambagaspitiya, S. (2026). Antioxidant and UV-protective potential of pigmented *Micrococcus* sp. isolated from *Padina* sp., collected in Hikkaduwa, Sri Lanka. Oral presentation at the *International Conference on Multidisciplinary Approaches (iCMA) 2026*, University of Sri Jayewardenepura, Sri Lanka.
7. Peiris, K.C.T., Dissanayake, I.A.J.K., Ariyawansa, K.G.S.U., & Kaumal, M.N. (2026). Extraction and characterization of a red pigment from *Talaromyces verruculosus* isolated from Kalpitiya Lagoon, Sri Lanka. Oral presentation at the *International Forestry and Environment Symposium 2026*, University of Sri Jayewardenepura, Sri Lanka.
8. Peiris, K.C.T., Kumanayaka, D.S., Deshani, N.A.D.K., Malka, U.G.S., Perera, K.L.S., Rathnasiri, H.A.D.G., Udeshika, L.A., Wimalarathna, W.G.N.A., Galagoda, R.R.P.W.M.R.U.C., Bandupriya, H.D.D., & Wickramasuriya, A.M. (2023, September). Evaluation of protocols for

the extraction of high-quality genomic DNA from *Piper nigrum* L. leaves. Oral presentation at the 43rd Annual Sessions of the Institute of Biology, Sri Lanka, Abstract p. 89.

RESEARCH AND TEACHING SKILLS

Laboratory Techniques

- Microbial isolation and fermentation
- DNA, RNA, and protein extraction
- PCR, Gel electrophoresis, and genetic transformation

Analytical Techniques

- TLC
- UV-Vis spectroscopy
- FTIR and NMR analysis

Data Analysis and Bioinformatics

- R, SPSS, MEGA
- Phylogenetic analysis
- Python (Basic)

Teaching and Academic Skills

- Undergraduate laboratory instruction
- Assessment and grading
- Scientific writing and presentations

HONORS & AWARDS

Dean's List

University of Colombo (2019/2020, 2021/2022, 2022/2023)

Recognized for academic excellence across three consecutive academic years.

PROFESSIONAL DEVELOPMENT & ONLINE CERTIFICATIONS

Selected professional development courses in genomics, bioinformatics, scientific computing, sustainability, and leadership completed through internationally recognized institutions, including Wellcome Connecting Science, RMIT University, and Deakin University.

VOLUNTEERING EXPERIENCE

1. Event host/moderator, The Botanical Society (BOTSOC) 2023 - 2024
Department of Plant Sciences, University of Colombo.

- Actively involved in organizing and leading seminars, Sci-Talk and Expertio series.

2. Co-author of "Research Spotlight" segment October - 2024
Botany Bloom, Department of Plant Sciences, University of Colombo.

3. Member, Organizing Committee, Sri Lankan Academy of Young Scientists (SLAYS) October - 2024

- Contributing to the 6th SLAYS Open Forum: "INDUSTRY 5.0 - Igniting Economic Resurgence of Sri Lanka".

CURRENT RESEARCH & RESEARCH INTEREST

Investigation of marine pigmented bacteria associated with brown seaweeds for antioxidant and UV-protective bioactive compounds, focusing on isolation, molecular characterization, fermentation optimisation and pigment analysis.

- Microbial secondary metabolites
- Natural pigments
- Marine microbiology
- Industrial biotechnology
- Metabolic engineering
- Natural product discovery
- Sustainable biomaterials

REFEREES

Dr. Jinendra Dissanayake, Ph.D

Senior Lecturer

Department of Plant Sciences

Faculty of Science

University of Colombo

Sri Lanka.

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Dr. K.G. Sameera U. Ariyawansa, Ph.D

Senior Lecturer

Department of Plant Sciences

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University of Colombo

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Dr. Isuru U. Kariyawasam, Ph.D

Senior Lecturer

Department of Botany

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